ELE 734 - Pre-Lab #2

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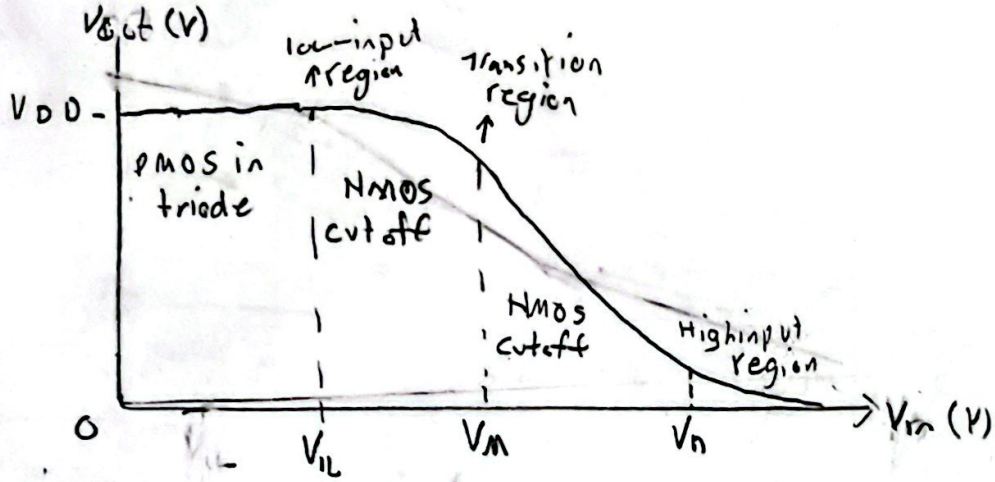
(1) The transfer characteristic of a CMOS inverter (V_{out} vs. V_{in}) shows output transitions from V_{DD} to low (0V) as input increases.
It has an "S"-shaped curve, with a sharp transition around V_M



Operating Regions:

$V_{in} = 0V \rightarrow$ NMOS OFF, PMOS ON $\rightarrow V_{out} \simeq V_{DD}$

$V_{in} \simeq V_{DD} \rightarrow$ NMOS ON, PMOS OFF $\rightarrow V_{out} \simeq 0V$

$V_{in} \simeq V_M \rightarrow$ transition Region

Increasing $\frac{\beta_p}{\beta_n}$ (stronger PMOS) $\rightarrow$ shifts V_M lower (left)

Decreasing $\frac{\beta_p}{\beta_n}$ (stronger NMOS) $\rightarrow$ shifts V_M higher (right)

(2) at $V_{out} = V_{in} = V_M$ (switching point), both transistors are in saturation:

$$I_{D,n} = \frac{1}{2}\beta_n (V_M - V_{Tn})^2$$

$$I_{D,p} = \frac{1}{2}\beta_p (V_{DD} - V_M - |V_{Tp}|)^2$$

at equilibrium ($I_{D,n} = I_{D,p}$);

$$\sqrt{\beta_n}(V_M - V_{Tn}) = \sqrt{\beta_p}(V_{DD} - V_M - |V_{Tp}|)$$

Solving for V_M;

$$\boxed{V_M = \frac{\sqrt{\beta_p}(V_{DD} - |V_{Tp}|) + \sqrt{\beta_n}\, V_{Tn}}{\sqrt{\beta_n} + \sqrt{\beta_p}}}$$

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(3) Noise margins:

$$NML = V_{IL} - V_{OL}, \quad NMH = V_{OH} - V_{IH}$$

when V_{DD} increases: (Unchanged V_T)

- V_{OH} decreases
- V_{IL} and V_{IH} move closer together
- NML and NMH shrink
- Lower noise immunity and unreliable switching

(4) $\tau_{pdr} = \ln(2) R_{eq,p} C_L, \quad \tau_{pdf} = \ln(2) R_{eq,n} C_L$

$$R_{eq} = \frac{V_{DD}}{2\ln(2) I_{D,sat}}, \quad I_{D,sat} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{DD} - V_T)^2$$

Solving for τ_p:

$$\boxed{\tau_p = \frac{C_L V_{DD}}{2}\left[\frac{1}{\mu_p C_{ox}\frac{W_p}{L_p}(V_{DD} - |V_{Tp}|)^2} + \frac{1}{\mu_n C_{ox}\frac{W_n}{L_n}(V_{DD} - V_{Tn})^2}\right]}$$

(5) Given:

- $\tau_p = 100ps$
- $C_L = 50fF$
- $V_{DD} = 1.8V$
- $L = 180nm$
- $\mu_n = 459 \frac{cm^2}{V \cdot s}$
- $\mu_p = 104 \frac{cm^2}{V \cdot s}$

$V_{Tn} = 0.445V$

$V_{Tp} = -0.438V$

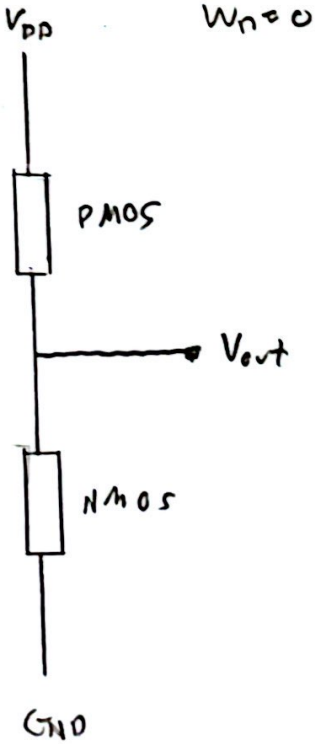
$t_{ox} = 4.08nm$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 8.46 \times 10^{-3} \frac{F}{m^2}$$

$$\frac{W_n}{L} = 1.26, \quad \frac{W_p}{L} = 5.26$$

$$\downarrow \qquad \downarrow$$

$$W_n = 0.23\mu m, \quad W_p = 0.95\mu m$$



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